The LOGISTIC Procedure

Model Information		
Data Set	WORK.PREDS	Predicted Values and Diagnostic Statistics
Response Variable	dfit	
Number of Response Levels	2	
Model	binary logit	
Optimization Technique	Fisher's scoring	

Number of Observations Read	139
Number of Observations Used	139

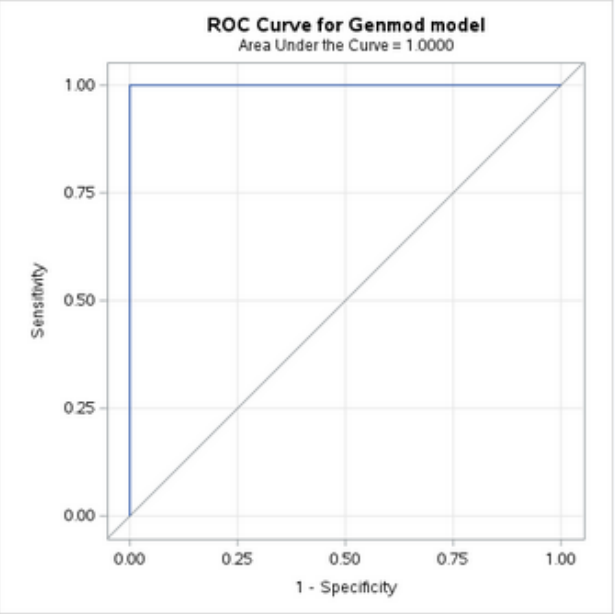
Response Profile		
Ordered Value	dfit	Total Frequency
1	1	5
2	0	134

Probability modeled is dfit=1.

Score Test for Global Null Hypothesis		
Chi-Square	DF	Pr > ChiSq
0.0000	0	<.0001

ROC Model: Genmod model

ROC Model Information		
ROC Contrast Coefficients	pred	Predicted Value



ROC Association Statistics							
	Mann-Whitney						
ROC Model	Area	Standard Error	95% Wald Confidence Limits		Somers' D	Gamma	Tau-a
Genmod model	1.0000	0	1.0000	1.0000	1.0000	1.0000	0.0699

The SURVEYSELECT Procedure

Selection Method	Simple Random Sampling

Input Data Set	DEFAULTED
Random Number Seed	965486363
Sample Size	5
Selection Probability	0.25
Sampling Weight	4
Output Data Set	DFLT_VALIDATION

The SURVEYSELECT Procedure

Selection Method	Simple Random Sampling

Input Data Set	NON_DEFAULTED_SELECT
Random Number Seed	965514596
Sample Size	5
Selection Probability	0.04065
Sampling Weight	24.6
Output Data Set	NON_DFLT_VALIDATION

The GENMOD Procedure

Model Information	
Data Set	WORK.MODEL_VALIDATION
Distribution	Binomial
Link Function	Logit
Dependent Variable	dflt
Scale Weight Variable	validation_sample

Number of Observations Read	143
Number of Observations Used	131
Sum of Weights	131
Number of Events	13
Number of Trials	131
Missing Values	12

Response Profile			
Ordered Value	dflt	Total Frequency	Total Weight
1	1	13	13
2	0	118	118

Criteria For Assessing Goodness Of Fit			
Criterion	DF	Value	Value/DF
Log Likelihood		-8.3481	
Full Log Likelihood		-8.3481	
AIC (smaller is better)		30.6961	
AICC (smaller is better)		31.6067	
BIC (smaller is better)		50.8225	

Algorithm converged.

Analysis Of Maximum Likelihood Parameter Estimates							
Parameter	DF	Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > Chi Sq
Intercept	1	-23.9848	7.7232	-39.1220	-8.8476	9.64	0.0019
Utilisation	1	9.7533	4.6752	0.5901	18.9165	4.35	0.0370
LTV	1	0.8136	0.5331	-0.2313	1.8586	2.33	0.1270
Borrowing_portfolio_	1	6.5154	4.7870	-2.8670	15.8977	1.85	0.1735
Postcode_index	1	0.0795	0.0507	-0.0199	0.1789	2.46	0.1168
arrears_flag	1	5.6209	2.3814	0.9535	10.2883	5.57	0.0183
relationship_length	1	0.1801	0.2741	-0.3572	0.7173	0.43	0.5112
Scale	0	1.0000	0.0000	1.0000	1.0000		

Note: The scale parameter was held fixed.

The MEANS Procedure

dflt=0

Variable	Label	N	Mean	Std Dev	Minimum	Maximum
Utilisation	Utilisation	123	0.5621138	0.2131160	0.0300000	0.8900000
LTV	LTV	123	2.0650402	1.3637526	0.0731707	6.5000000
Borrowing_portfolio_ratio	Borrowing_portfolio_ratio	123	0.1288171	0.1058244	0.0010996	0.5503003
Postcode_index	Postcode_index	123	73.6910569	14.0497536	50.0000000	99.0000000
arrears_flag		123	0.1626016	0.3705110	0	1.0000000
relationship_length		123	5.7235772	3.0922982	1.0000000	9.0000000

dflt=1

Variable	Label	N	Mean	Std Dev	Minimum	Maximum
Utilisation	Utilisation	20	0.9265674	0.2727838	0.5300000	1.8933874
LTV	LTV	20	9.2794160	17.1847257	0.2445092	67.1443259
Borrowing_portfolio_ratio	Borrowing_portfolio_ratio	20	0.4070063	0.2427159	0.1056738	0.8840264
Postcode_index	Postcode_index	20	82.3000000	11.7791073	55.0000000	98.0000000
arrears_flag		20	0.8000000	0.4103913	0	1.0000000
relationship_length		20	7.4500000	1.9861362	2.0000000	9.0000000

Customers in validaiton dflt selection

Customer
1487640
3342349
3779903
7923120
35234232

Customers in validaiton non dflt selection

Customer
2273231
3405484
4264564
17238907
18308234

Prediction of validation dataset

Customer	observed_default_status	Predicted Value
1487640	1	1.00000
2273231	0	0.00000
3342349	1	0.07892
3405484	0	0.02905
3779903	1	0.99340
3779903	1	0.60943
4264564	0	0.06055
7923120	1	0.99931
17238907	0	0.00039
18308234	0	0.00375
35234232	1	0.87226
35234232	1	0.55847

Prediction of validation dataset

The LOGISTIC Procedure

Model Information		
Data Set	WORK.PREDS	Predicted Values and Diagnostic Statistics
Response Variable	dfit	
Number of Response Levels	2	
Model	binary logit	
Optimization Technique	Fisher's scoring	

Number of Observations Read	12
Number of Observations Used	12

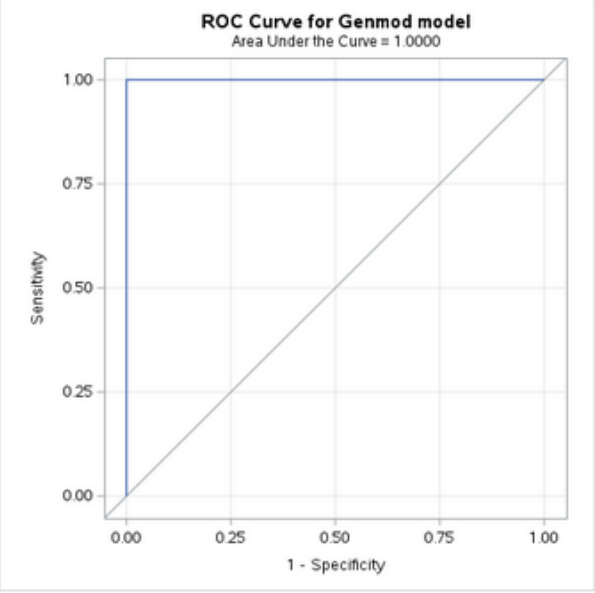
Response Profile		
Ordered Value	dfit	Total Frequency
1	1	7
2	0	5

Probability modeled is dfit=1.

Score Test for Global Null Hypothesis		
Chi-Square	DF	Pr > ChiSq
0.0000	0	<.0001

ROC Model: Genmod model

ROC Model Information		
ROC Contrast Coefficients	pred	Predicted Value



ROC Association Statistics							
	Mann-Whitney						
ROC Model	Area	Standard Error	95% Wald Confidence Limits		Somers' D	Gamma	Tau-a
Genmod model	1.0000	0	1.0000	1.0000	1.0000	1.0000	0.5303

Prediction of validation dataset

The LOGISTIC Procedure

Model Information	
Data Set	WORK.LOGISTIC
Response Variable	dflt
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	132
Number of Observations Used	132

Response Profile		
Ordered Value	dflt	Total Frequency
1	0	118
2	1	14

Probability modeled is dflt=0.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	91.285	31.122
SC	94.167	51.302
-2 Log L	89.285	17.122

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	72.1625	6	<.0001
Score	78.7635	6	<.0001
Wald	9.3022	6	0.1573

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	24.2386	8.0849	9.0327	0.0027
Utilization	1	-9.2090	4.8407	3.6191	0.0571
LTV	1	-0.8409	0.5561	2.2670	0.1305
Borrowing_portfolio	1	-6.2435	4.7283	1.7436	0.1867
Postcode_index	1	-0.0789	0.0505	2.4391	0.1183
arrears_flag	1	-5.9412	2.5582	5.3937	0.0202
relationship_length	1	-0.2809	0.2891	0.8145	0.3668

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
Utilization	<0.001	<0.001	1.321
LTV	0.431	0.145	1.283
Borrowing_portfolio	0.002	<0.001	20.568
Postcode_index	0.924	0.837	1.020
arrears_flag	0.003	<0.001	0.396
relationship_length	0.770	0.437	1.358

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	98.8	Somers' D	0.977
Percent Discordant	1.2	Gamma	0.977
Percent Tied	0.0	Tau-a	0.187
Pairs	1652	c	0.988

Prediction of validation dataset

The GENMOD Procedure

Model Information	
Data Set	WORK.OVERALL
Distribution	Binomial
Link Function	Probit
Dependent Variable	diff
Scale Weight Variable	validation_sample

Number of Observations Read	143
Number of Observations Used	132
Sum of Weights	132
Number of Events	14
Number of Trials	132
Missing Values	11

Response Profile			
Ordered Value	diff	Total Frequency	Total Weight
1	1	14	14
2	0	118	118

Criteria For Assessing Goodness Of Fit			
Criterion	DF	Value	Value/DF
Log Likelihood		-11.1309	
Full Log Likelihood		-11.1309	
AIC (smaller is better)		32.2619	
AICC (smaller is better)		32.7381	
BIC (smaller is better)		46.6759	

Algorithm converged.

Analysis Of Maximum Likelihood Parameter Estimates							
Parameter	DF	Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > ChiSq
Intercept	1	-9.9197	2.8782	-15.1689	-4.6705	13.72	0.0002
Utilisation	1	6.2753	2.2933	1.7808	10.7700	7.49	0.0062
Borrowing_portfolio	1	3.8592	1.9509	0.0355	7.6829	3.91	0.0479
Postcode_index	1	0.0265	0.0221	-0.0168	0.0698	1.44	0.2302
arrears_flag	1	2.3829	0.7584	0.8805	3.8454	9.78	0.0018
Scale	0	1.0000	0.0000	1.0000	1.0000		

Note: The scale parameter was held fixed.

Prediction of validation dataset

Parameter	Estimate
Intercept	-9.919741832
Utilisation	6.275318844
Borrowing_portfolio_	3.859191182
Postcode_index	0.026509379
arrears_flag	2.382919789
Scale	1.000000000

Prediction of validation dataset

The LOGISTIC Procedure

Model Information		
Data Set	WORK.PREDS	Predicted Values and Diagnostic Statistics
Response Variable	dflt	
Number of Response Levels	2	
Model	binary logit	
Optimization Technique	Fisher's scoring	

Number of Observations Read	11
Number of Observations Used	11

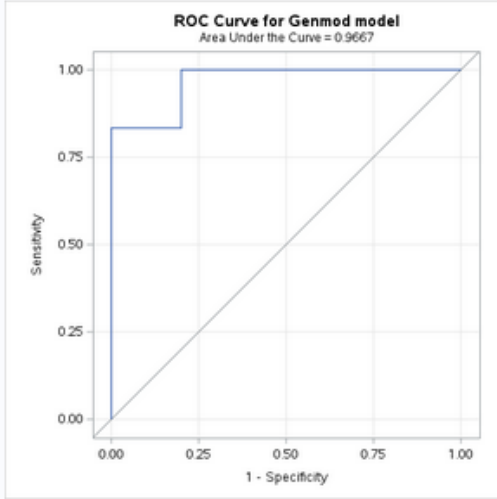
Response Profile		
Ordered Value	dflt	Total Frequency
1	1	6
2	0	5

Probability modeled is dflt=1.

Score Test for Global Null Hypothesis		
Chi-Square	DF	Pr > ChiSq
0.0000	0	<.0001

ROC Model: Genmod model

ROC Model Information		
ROC Contrast Coefficients	pred	Predicted Value



ROC Association Statistics							
	Mann-Whitney						
ROC Model	Area	Standard Error	95% Wald Confidence Limits		Somers' D	Gamma	Tau-a
Genmod model	0.9667	0.0471	0.8743	1.0000	0.9333	0.9333	0.5091

Prediction of validation dataset

The CORR Procedure

6 Variables:	Utilisation LTV Borrowing_portfolio_ratio Postcode_index arrears_flag relationship_length

Simple Statistics							
Variable	N	Mean	Std Dev	Sum	Minimum	Maximum	Label
Utilisation	143	0.61309	0.25509	87.67135	0.03000	1.89339	Utilisation
LTV	143	3.07404	6.88801	439.58828	0.07317	67.14433	LTV
Borrowing_portfolio_ratio	143	0.18772	0.16395	26.86463	0.00110	0.88403	Borrowing_portfolio_ratio
Postcode_index	143	74.89510	14.04054	10710	50.00000	99.00000	Postcode_index
arrears_flag	143	0.25175	0.43554	36.00000	0	1.00000	
relationship_length	143	5.98503	3.01735	855.60000	1.00000	9.00000	

Pearson Correlation Coefficients, N = 143 Prob > |r| under H0: Rho=0						
	Utilisation	LTV	Borrowing_portfolio_ratio	Postcode_index	arrears_flag	relationship_length
Utilisation Utilisation	1.00000	0.28092 0.0007	0.43981 <.0001	0.02092 0.8041	0.20832 0.0125	0.11063 0.1884
LTV LTV	0.28092 0.0007	1.00000	0.22750 0.0063	-0.02823 0.7379	0.06764 0.4222	0.04857 0.5808
Borrowing_portfolio_ratio Borrowing_portfolio_ratio	0.43981 <.0001	0.22750 0.0063	1.00000	0.05569 0.5088	0.30887 0.0002	-0.02854 0.7351
Postcode_index Postcode_index	0.02092 0.8041	-0.02823 0.7379	0.05569 0.5088	1.00000	-0.00602 0.9432	0.07488 0.3741
arrears_flag	0.20832 0.0125	0.06764 0.4222	0.30887 0.0002	-0.00602 0.9432	1.00000	0.06033 0.4741
relationship_length	0.11063 0.1884	0.04857 0.5808	-0.02854 0.7351	0.07488 0.3741	0.06033 0.4741	1.00000

Total rows: 20 Total columns: 18

Rows 1-20

	Customer	Year	Utilisation	Limit	Borrowing	LTV	Collateral	Collateral_type	Portfolio_value	Borro
1	1487640	2017	0.99	507,936	500317.44	50.031744	10000	3	1,927,484	
2	2501838	2015	0.57	1,621,391	924278.2	3.2270334229	286417.3	2	2,340,989	
3	3342349	2016	0.63	1,693,786	1067085.18	4.1329710398	258188.4	3	1,693,786	
4	3779903	2010	0.93	1,544,890	1435076.3	3.7683236483	380826.18	2	2,444,407	
5	3779903	2015	0.74	1,061,059	789688.35	2.4808182203	318317.7	2	2,444,407	
6	3982426	2017	1.00	802,703	802027.25	6.8708006121	116729.81	2	5,326,520	
7	5182106	2014	1.02	1,324,381	1354004	7.2999700885	185480.76	2	1,930,781	
8	6161840	2016	0.98	1,385,358	1358903	3.1383441109	433000	4	1,839,078	
9	7923120	2016	1.08	1,136,257	1224941.35	3.9927761191	306789.39	3	2,169,741	
10	8064461	2014	0.91	1,801,127	1630394.45	5.028932965	324202.86	3	1,844,283	
11	8141414	2011	0.53	650,088	344546.64	2.0507752012	168008	3	1,999,912	
12	8546204	2010	0.91	299,686	273817.88	3.1115668182	88000	2	1,993,292	
13	8546204	2015	1.89	152,278	288321.24	7.0827826137	40707.34	2	1,993,292	
14	8697888	2012	0.91	1,927,963	1753175.95	67.144325897	26110.56	3	3,855,926	
15	12095275	2010	0.90	1,813,476	1623837.48	3.3637965818	482739.5	2	8,816,373	
16	12322385	2014	0.89	899,853	803917.68	4.9945999987	160957.37	3	1,069,662	
17	19992073	2010	1.03	264,303	271796.95	0.244509178	1111602.24	2	2,572,038	
18	19992073	2013	0.95	1,089,844	1030195.48	3.6356490541	283359.44	2	2,572,038	
19	35234232	2012	0.86	742,408	638020.72	2.5514051247	250066.41	3	2,075,504	
20	35234232	2017	0.82	628,136	514569.76	1.4371950739	358037.52	3	2,075,504	